

Project Title

LED Control with Voice Commands

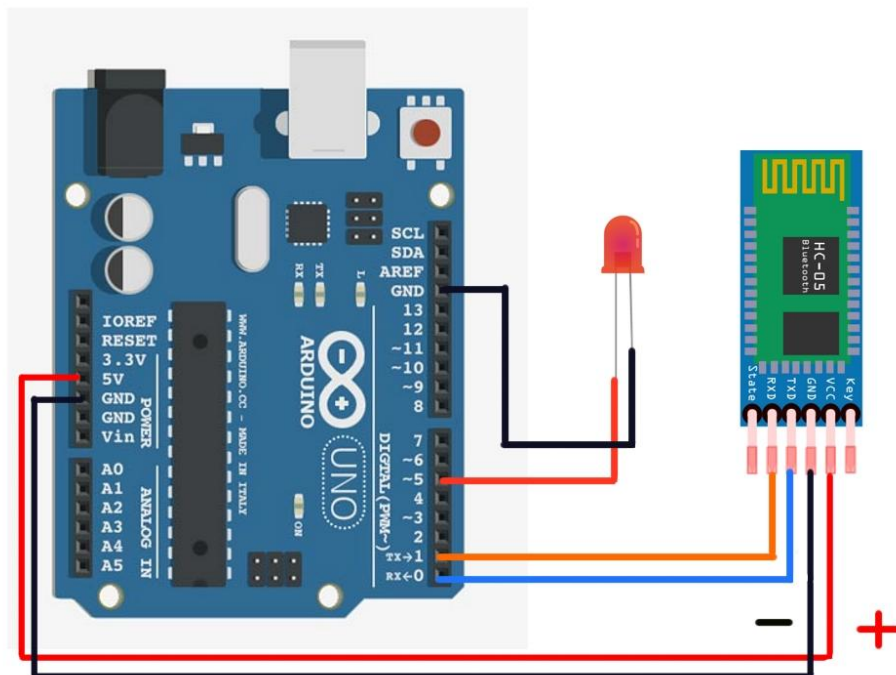
Introduction

This project demonstrates how to control LEDs using voice commands through a serial interface. The implementation uses an Arduino board, a few LEDs, and a serial monitor to receive and process voice commands. The project aims to provide an interactive way to control LEDs with simple voice commands.

Components

- Arduino board (e.g., Arduino Uno)
- LEDs (Green, blue, Red)
- Resistors (appropriate value for LEDs)
- Breadboard and jumper wires
- Serial monitor for communication

Circuit Diagram



Code Explanation

```
1  int Green = 5;
2  int Blue = 4;
3  int Red = 3;
4  bool blinkGreen = false;
5  bool blinkBlue = false;
6  bool blinkRed = false;
7  unsigned long previousMillis = 0;
8  const long interval = 500; // Interval at which to blink (milliseconds)
9
10 void setup() {
11     Serial.begin(9600);
12     pinMode(Green, OUTPUT);
13     pinMode(Blue, OUTPUT);
14     pinMode(Red, OUTPUT);
15 }
16 |
17 String voice;
18
19 void loop() {
20     if (Serial.available() > 0) {
21         voice = Serial.readString();
22         voice.trim(); // Trim any leading/trailing whitespace
23         Serial.print(voice + '\n');
24
25         if (voice.equalsIgnoreCase("green")) {
26             digitalWrite(Green, HIGH);
27             blinkGreen = false;
28         } else if (voice.equalsIgnoreCase("green off")) {
29             digitalWrite(Green, LOW);
30             blinkGreen = false;
31         }
32
33         if (voice.equalsIgnoreCase("blue")) {
```

```
33     if (voice.equalsIgnoreCase("blue")) {
34         digitalWrite(Blue, HIGH);
35         blinkBlue = false;
36     } else if (voice.equalsIgnoreCase("blue off")) {
37         digitalWrite(Blue, LOW);
38         blinkBlue = false;
39     }
40
41     if (voice.equalsIgnoreCase("red")) {
42         digitalWrite(Red, HIGH);
43         blinkRed = false;
44     } else if (voice.equalsIgnoreCase("red off")) {
45         digitalWrite(Red, LOW);
46         blinkRed = false;
47     }
48
49     if (voice.equalsIgnoreCase("all lights on")) {
50         digitalWrite(Green, HIGH);
51         digitalWrite(Blue, HIGH);
52         digitalWrite(Red, HIGH);
53         blinkGreen = blinkBlue = blinkRed = false;
54     } else if (voice.equalsIgnoreCase("all lights off")) {
55         digitalWrite(Green, LOW);
56         digitalWrite(Blue, LOW);
57         digitalWrite(Red, LOW);
58         blinkGreen = blinkBlue = blinkRed = false;
59     }
60
61     if (voice.equalsIgnoreCase("blink green")) {
62         blinkGreen = true;
63     } else if (voice.equalsIgnoreCase("blink blue")) {
64         blinkBlue = true;
65     } else if (voice.equalsIgnoreCase("blink red")) {
```

```
65     } else if (voice.equalsIgnoreCase("blink red")) {
66         blinkRed = true;
67     }
68
69     if (voice.equalsIgnoreCase("turn off green")) {
70         digitalWrite(Green, LOW);
71         blinkGreen = false;
72     } else if (voice.equalsIgnoreCase("turn off blue")) {
73         digitalWrite(Blue, LOW);
74         blinkBlue = false;
75     } else if (voice.equalsIgnoreCase("turn off red")) {
76         digitalWrite(Red, LOW);
77         blinkRed = false;
78     }
79 }
80
81 unsigned long currentMillis = millis();
82
83 if (currentMillis - previousMillis >= interval) {
84     previousMillis = currentMillis;
85
86     if (blinkGreen) {
87         int state = digitalRead(Green);
88         digitalWrite(Green, !state);
89     }
90
91     if (blinkBlue) {
92         int state = digitalRead(Blue);
93         digitalWrite(Blue, !state);
94     }
95
96     if (blinkRed) {
```

```
96         if (blinkRed) {
97             int state = digitalRead(Red);
98             digitalWrite(Red, !state);
99         }
100     }
101 }
102
```

Functionality

Voice Commands:

- **"green"**: Turns on the green LED.
- **"turn off green"**: Turns off the green LED.
- **"blue"**: Turns on the blue LED.
- **"turn off blue"**: Turns off the blue LED.
- **"red"**: Turns on the red LED.
- **"turn off red "**: Turns off the red LED.
- **"all lights on"**: Turns on all LEDs.
- **"all lights off"**: Turns off all LEDs.
- **"blink green"**: Makes the green LED blink.
- **"blink blue"**: Makes the blue LED blink.
- **"blink red"**: Makes the red LED blink.

Blinking Mechanism:

The blinking functionality is implemented using the `millis()` function to toggle the LED states at defined intervals.

Conclusion

This project successfully demonstrates the use of voice commands to control LEDs, showcasing the integration of serial communication and LED control. It provides a practical application of Arduino in creating interactive electronic projects.

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